LABSHEET-11

Java GUI & Event Handling

Q1. Student Registration Form - AWT

Create a Java AWT application to design a Student Registration Form containing Label and TextField for Student Name, Roll Number and Course; Checkbox for Gender; Choice for Semester; Submit and Reset buttons. Submit should display entered information; Reset should clear all fields.

```
import java.awt.*;
import java.awt.event.*;

public class Main extends Frame implements ActionListener {
    Label lName, lRoll, lCourse, lGender, lSem;
    TextField tName, tRoll, tCourse;
    Checkbox male, female, other;
    CheckboxGroup genderGroup;
    Choice semester;
    Button submit, reset;
    TextArea output;

    Main() {
        setTitle("Student Registration Form");
        setLayout(new BorderLayout(10, 10));

        Panel form = new Panel(new GridLayout(6, 2, 10, 10));

        lName = new Label("Student Name:");
        tName = new TextField(20);
        lRoll = new Label("Roll Number:");
        tRoll = new TextField(20);
        lCourse = new Label("Course:");
        tCourse = new TextField(20);

        lGender = new Label("Gender:");
        genderGroup = new CheckboxGroup();
        male = new Checkbox("Male", genderGroup, false);
        female = new Checkbox("Female", genderGroup, false);
        other = new Checkbox("Other", genderGroup, false);
        Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
        genderPanel.add(male);
        genderPanel.add(female);
        genderPanel.add(other);

        lSem = new Label("Semester:");
        semester = new Choice();
        for (int i = 1; i <= 8; i++) {
            semester.add("Semester " + i);
        }

        submit = new Button("Submit");
        reset = new Button("Reset");

        form.add(lName);   form.add(tName);
        form.add(lRoll);   form.add(tRoll);
        form.add(lCourse); form.add(tCourse);
        form.add(lGender); form.add(genderPanel);
        form.add(lSem);    form.add(semester);
        form.add(submit);  form.add(reset);

        output = new TextArea(6, 30);
        output.setEditable(false);

        add(new Label("   "), BorderLayout.WEST);
        add(form, BorderLayout.CENTER);
        add(output, BorderLayout.SOUTH);

        submit.addActionListener(this);
        reset.addActionListener(this);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });
```

```
        setSize(430, 470);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == submit) {
            Checkbox selected = genderGroup.getSelectedCheckbox();
            String gender = (selected == null) ? "Not selected" : selected.getLabel();
            output.setText("Student Name : " + tName.getText() + "\n"
                    + "Roll Number  : " + tRoll.getText() + "\n"
                    + "Course       : " + tCourse.getText() + "\n"
                    + "Gender       : " + gender + "\n"
                    + "Semester     : " + semester.getSelectedItem());
        } else if (e.getSource() == reset) {
            tName.setText("");
            tRoll.setText("");
            tCourse.setText("");
            genderGroup.setSelectedCheckbox(null);
            semester.select(0);
            output.setText("");
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Input given in the form:
Student Name : Varnika | Roll Number : CU25260192 | Course : BCA
Gender : Female | Semester : Semester 2

On clicking Submit, the TextArea shows:
Student Name : Varnika
Roll Number  : CU25260192
Course       : BCA
Gender       : Female
Semester     : Semester 2

On clicking Reset: all text fields, the gender selection and the
TextArea are cleared and Semester goes back to "Semester 1".
```

Q2. Simple Calculator - AWT

Create a Java AWT-based calculator with two TextFields, four Buttons for Addition, Subtraction, Multiplication and Division, and a result display. Perform the selected operation and handle division by zero.

```
import java.awt.*;
import java.awt.event.*;

public class Main extends Frame implements ActionListener {
    Label l1, l2, lRes;
    TextField t1, t2, result;
    Button add, sub, mul, div;

    Main() {
        setTitle("Simple Calculator");
        setLayout(new GridLayout(5, 2, 10, 10));

        l1 = new Label("First Number:");
        t1 = new TextField();
        l2 = new Label("Second Number:");
        t2 = new TextField();
        add = new Button("Addition");
        sub = new Button("Subtraction");
        mul = new Button("Multiplication");
        div = new Button("Division");
        lRes = new Label("Result:");
        result = new TextField();
        result.setEditable(false);

        add(l1);  add(t1);
        add(l2);  add(t2);
        add(add); add(sub);
        add(mul); add(div);
        add(lRes); add(result);

        add.addActionListener(this);
```

```
        sub.addActionListener(this);
        mul.addActionListener(this);
        div.addActionListener(this);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });

        setSize(380, 280);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        double a, b;
        try {
            a = Double.parseDouble(t1.getText().trim());
            b = Double.parseDouble(t2.getText().trim());
        } catch (NumberFormatException ex) {
            result.setText("Invalid input");
            return;
        }

        if (e.getSource() == add) {
            result.setText(String.valueOf(a + b));
        } else if (e.getSource() == sub) {
            result.setText(String.valueOf(a - b));
        } else if (e.getSource() == mul) {
            result.setText(String.valueOf(a * b));
        } else if (e.getSource() == div) {
            if (b == 0) {
                result.setText("Cannot divide by zero");
            } else {
                result.setText(String.valueOf(a / b));
            }
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
First Number = 20, Second Number = 5
Addition       -> Result: 25.0
Subtraction    -> Result: 15.0
Multiplication -> Result: 100.0
Division       -> Result: 4.0

First Number = 20, Second Number = 0
Division       -> Result: Cannot divide by zero

If a field is empty or not a number:
Result: Invalid input
```

Q3. Student Login Form - Swing

Create a Swing login form with JLabel/JTextField for username, JLabel/JPasswordField for password, Login and Reset buttons. On Login, compare with predefined credentials and display the result using JOptionPane.

```
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;
import java.awt.event.*;

public class Main extends JFrame implements ActionListener {
    JLabel lUser, lPass;
    JTextField tUser;
    JPasswordField tPass;
    JButton login, reset;

    // predefined credentials
    final String USERNAME = "varnika";
    final String PASSWORD = "cu123";

    Main() {
        setTitle("Student Login");
```

```
        JPanel panel = new JPanel(new GridLayout(3, 2, 10, 15));
        panel.setBorder(new EmptyBorder(20, 20, 20, 20));

        lUser = new JLabel("Username:");
        tUser = new JTextField(15);
        lPass = new JLabel("Password:");
        tPass = new JPasswordField(15);
        login = new JButton("Login");
        reset = new JButton("Reset");

        panel.add(lUser);  panel.add(tUser);
        panel.add(lPass);  panel.add(tPass);
        panel.add(login);  panel.add(reset);

        login.addActionListener(this);
        reset.addActionListener(this);

        add(panel);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(350, 220);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == login) {
            String user = tUser.getText().trim();
            String pass = new String(tPass.getPassword());

            if (user.isEmpty() || pass.isEmpty()) {
                JOptionPane.showMessageDialog(this, "Please enter username and password.",
                        "Warning", JOptionPane.WARNING_MESSAGE);
            } else if (user.equals(USERNAME) && pass.equals(PASSWORD)) {
                JOptionPane.showMessageDialog(this, "Login Successful! Welcome, " + user + ".",
                        "Login", JOptionPane.INFORMATION_MESSAGE);
            } else {
                JOptionPane.showMessageDialog(this, "Invalid username or password.",
                        "Login Failed", JOptionPane.ERROR_MESSAGE);
            }
        } else if (e.getSource() == reset) {
            tUser.setText("");
            tPass.setText("");
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Predefined credentials: username = varnika, password = cu123

Username: varnika, Password: cu123, click Login
JOptionPane message: Login Successful! Welcome, varnika.

Username: varnika, Password: wrong, click Login
JOptionPane message: Invalid username or password.

Username or Password left empty, click Login
JOptionPane message: Please enter username and password.

On clicking Reset: username and password fields are cleared.
```

Q4. Student Marksheet GUI - Swing

Create a Swing application accepting marks for English, Mathematics, Java, DBMS and Computer Networks. Add a Calculate Result button to display total, percentage, grade and pass/fail status.

```
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;
import java.awt.event.*;

public class Main extends JFrame implements ActionListener {
    String[] subjects = {"English", "Mathematics", "Java", "DBMS", "Computer Networks"};
    JTextField[] marks = new JTextField[5];
    JButton calculate;
    JTextArea result;
```

```
    Main() {
        setTitle("Student Marksheet");
        setLayout(new BorderLayout(10, 10));

        JPanel panel = new JPanel(new GridLayout(6, 2, 10, 10));
        panel.setBorder(new EmptyBorder(15, 20, 5, 20));

        for (int i = 0; i < 5; i++) {
            panel.add(new JLabel(subjects[i] + " (out of 100):"));
            marks[i] = new JTextField();
            panel.add(marks[i]);
        }
        calculate = new JButton("Calculate Result");
        panel.add(new JLabel(""));
        panel.add(calculate);
        calculate.addActionListener(this);

        result = new JTextArea(6, 25);
        result.setEditable(false);
        result.setBorder(new EmptyBorder(8, 20, 8, 20));

        add(panel, BorderLayout.CENTER);
        add(result, BorderLayout.SOUTH);

        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(420, 450);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        int total = 0;
        boolean pass = true;
        try {
            for (int i = 0; i < 5; i++) {
                int m = Integer.parseInt(marks[i].getText().trim());
                if (m < 0 || m > 100) {
                    JOptionPane.showMessageDialog(this, "Marks must be between 0 and 100.");
                    return;
                }
                total += m;
                if (m < 40) {
                    pass = false;
                }
            }
        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(this, "Please enter valid marks in all fields.");
            return;
        }

        double percentage = total / 5.0;
        String grade;
        if (!pass) grade = "F";
        else if (percentage >= 90) grade = "A+";
        else if (percentage >= 80) grade = "A";
        else if (percentage >= 70) grade = "B+";
        else if (percentage >= 60) grade = "B";
        else if (percentage >= 50) grade = "C";
        else grade = "D";

        result.setText("Total Marks : " + total + " / 500\n"
                + "Percentage  : " + String.format("%.2f", percentage) + " %\n"
                + "Grade       : " + grade + "\n"
                + "Status      : " + (pass ? "PASS" : "FAIL"));
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Marks: English 78, Mathematics 85, Java 92, DBMS 67, Computer Networks 88
On clicking Calculate Result:
Total Marks : 410 / 500
Percentage  : 82.00 %
Grade       : A
Status      : PASS

Marks: English 78, Mathematics 85, Java 92, DBMS 35, Computer Networks 88
Total Marks : 378 / 500
Percentage  : 75.60 %
Grade       : F
Status      : FAIL      (a subject below 40 marks)

Invalid or empty marks -> "Please enter valid marks in all fields."
Marks outside 0-100    -> "Marks must be between 0 and 100."
```

Q5. Course Selection Form - Swing

Create a Swing GUI with Student Name (JTextField), Course (JComboBox), Gender (JRadioButton), Subjects (multiple JCheckBox), Submit and Reset buttons. Submit should display all selected information using JOptionPane.

```
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;
import java.awt.event.*;

public class Main extends JFrame implements ActionListener {
    JTextField tName;
    JComboBox<String> course;
    JRadioButton male, female, other;
    ButtonGroup genderGroup;
    JCheckBox[] subjects = new JCheckBox[4];
    JButton submit, reset;

    Main() {
        setTitle("Course Selection Form");

        JPanel panel = new JPanel(new GridLayout(6, 2, 10, 12));
        panel.setBorder(new EmptyBorder(15, 20, 15, 20));

        tName = new JTextField();
        course = new JComboBox<>(new String[]{"BCA", "BBA", "BTech", "MCA", "MBA"});

        male = new JRadioButton("Male");
        female = new JRadioButton("Female");
        other = new JRadioButton("Other");
        genderGroup = new ButtonGroup();
        genderGroup.add(male);
        genderGroup.add(female);
        genderGroup.add(other);
        JPanel genderPanel = new JPanel(new FlowLayout(FlowLayout.LEFT, 5, 0));
        genderPanel.add(male);
        genderPanel.add(female);
        genderPanel.add(other);

        String[] names = {"Java", "DBMS", "Mathematics", "Computer Networks"};
        JPanel subjectPanel = new JPanel(new GridLayout(2, 2));
        for (int i = 0; i < 4; i++) {
            subjects[i] = new JCheckBox(names[i]);
            subjectPanel.add(subjects[i]);
        }

        submit = new JButton("Submit");
        reset = new JButton("Reset");

        panel.add(new JLabel("Student Name:")); panel.add(tName);
        panel.add(new JLabel("Course:"));       panel.add(course);
        panel.add(new JLabel("Gender:"));       panel.add(genderPanel);
        panel.add(new JLabel("Subjects:"));     panel.add(subjectPanel);
        panel.add(new JLabel(""));              panel.add(new JLabel(""));
        panel.add(submit);                      panel.add(reset);

        submit.addActionListener(this);
        reset.addActionListener(this);

        add(panel);
```

```
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(450, 340);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == submit) {
            String gender = "Not selected";
            if (male.isSelected()) gender = "Male";
            else if (female.isSelected()) gender = "Female";
            else if (other.isSelected()) gender = "Other";

            String selected = "";
            for (JCheckBox cb : subjects) {
                if (cb.isSelected()) {
                    selected += cb.getText() + ", ";
                }
            }
            if (selected.isEmpty()) {
                selected = "None";
            } else {
                selected = selected.substring(0, selected.length() - 2);
            }

            JOptionPane.showMessageDialog(this,
                    "Student Name : " + tName.getText() + "\n"
                    + "Course       : " + course.getSelectedItem() + "\n"
                    + "Gender       : " + gender + "\n"
                    + "Subjects     : " + selected,
                    "Selected Information", JOptionPane.INFORMATION_MESSAGE);
        } else if (e.getSource() == reset) {
            tName.setText("");
            course.setSelectedIndex(0);
            genderGroup.clearSelection();
            for (JCheckBox cb : subjects) {
                cb.setSelected(false);
            }
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Input: Name = Varnika, Course = BCA, Gender = Female,
Subjects = Java and DBMS selected

On clicking Submit, JOptionPane shows (title: Selected Information):
Student Name : Varnika
Course       : BCA
Gender       : Female
Subjects     : Java, DBMS

On clicking Reset: name is cleared, course goes back to BCA, gender and
subject selections are removed.
```

Q6. Employee Registration Form - AWT

Create an AWT Employee Registration Form with Employee ID, Name, Department, Designation, Salary, Gender, Department selection, Submit and Clear buttons. Display submitted information in a TextArea and clear fields using Clear.

```
import java.awt.*;
import java.awt.event.*;

public class Main extends Frame implements ActionListener {
    TextField tId, tName, tDesignation, tSalary;
    CheckboxGroup genderGroup;
    Checkbox male, female;
    Choice department;
    Button submit, clear;
    TextArea output;

    Main() {
        setTitle("Employee Registration Form");
        setLayout(new BorderLayout(10, 10));
```

```
        Panel form = new Panel(new GridLayout(7, 2, 10, 10));

        tId = new TextField();
        tName = new TextField();
        tDesignation = new TextField();
        tSalary = new TextField();

        genderGroup = new CheckboxGroup();
        male = new Checkbox("Male", genderGroup, false);
        female = new Checkbox("Female", genderGroup, false);
        Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
        genderPanel.add(male);
        genderPanel.add(female);

        department = new Choice();
        department.add("Computer Science");
        department.add("Information Technology");
        department.add("Human Resources");
        department.add("Finance");
        department.add("Sales");

        submit = new Button("Submit");
        clear = new Button("Clear");

        form.add(new Label("Employee ID:"));   form.add(tId);
        form.add(new Label("Name:"));          form.add(tName);
        form.add(new Label("Designation:"));   form.add(tDesignation);
        form.add(new Label("Salary:"));        form.add(tSalary);
        form.add(new Label("Gender:"));        form.add(genderPanel);
        form.add(new Label("Department:"));    form.add(department);
        form.add(submit);                      form.add(clear);

        output = new TextArea(7, 30);
        output.setEditable(false);

        add(new Label("   "), BorderLayout.WEST);
        add(form, BorderLayout.CENTER);
        add(output, BorderLayout.SOUTH);

        submit.addActionListener(this);
        clear.addActionListener(this);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });

        setSize(440, 540);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == submit) {
            Checkbox sel = genderGroup.getSelectedCheckbox();
            String gender = (sel == null) ? "Not selected" : sel.getLabel();
            output.setText("Employee ID : " + tId.getText() + "\n"
                    + "Name        : " + tName.getText() + "\n"
                    + "Designation : " + tDesignation.getText() + "\n"
                    + "Salary      : " + tSalary.getText() + "\n"
                    + "Gender      : " + gender + "\n"
                    + "Department  : " + department.getSelectedItem());
        } else if (e.getSource() == clear) {
            tId.setText("");
            tName.setText("");
            tDesignation.setText("");
            tSalary.setText("");
            genderGroup.setSelectedCheckbox(null);
            department.select(0);
            output.setText("");
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Input: Employee ID = E101, Name = Varnika, Designation = Software Developer,
Salary = 55000, Gender = Female, Department = Computer Science

On clicking Submit, the TextArea shows:
Employee ID : E101
Name        : Varnika
Designation : Software Developer
Salary      : 55000
Gender      : Female
Department  : Computer Science

On clicking Clear: all fields, gender and the TextArea are cleared and
Department goes back to "Computer Science".
```

Q7. Temperature Converter - Swing

Create a Swing temperature converter with a temperature TextField, Celsius-to-Fahrenheit and Fahrenheit-to-Celsius radio buttons, Convert and Clear buttons, and result display. Use F=(C*9/5)+32 and C=(F-32)*5/9. Handle invalid input.

```
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;
import java.awt.event.*;

public class Main extends JFrame implements ActionListener {
    JTextField tTemp;
    JRadioButton cToF, fToC;
    ButtonGroup group;
    JButton convert, clear;
    JLabel result;

    Main() {
        setTitle("Temperature Converter");

        JPanel panel = new JPanel(new GridLayout(4, 1, 8, 12));
        panel.setBorder(new EmptyBorder(15, 20, 15, 20));

        JPanel p1 = new JPanel(new FlowLayout());
        p1.add(new JLabel("Temperature:"));
        tTemp = new JTextField(12);
        p1.add(tTemp);

        cToF = new JRadioButton("Celsius to Fahrenheit", true);
        fToC = new JRadioButton("Fahrenheit to Celsius");
        group = new ButtonGroup();
        group.add(cToF);
        group.add(fToC);
        JPanel p2 = new JPanel(new FlowLayout());
        p2.add(cToF);
        p2.add(fToC);

        convert = new JButton("Convert");
        clear = new JButton("Clear");
        JPanel p3 = new JPanel(new FlowLayout());
        p3.add(convert);
        p3.add(clear);

        result = new JLabel("Result: ", SwingConstants.CENTER);
        result.setFont(new Font("Arial", Font.BOLD, 14));

        panel.add(p1);
        panel.add(p2);
        panel.add(p3);
        panel.add(result);

        convert.addActionListener(this);
        clear.addActionListener(this);

        add(panel);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(430, 270);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
```

```
        if (e.getSource() == convert) {
            try {
                double temp = Double.parseDouble(tTemp.getText().trim());
                if (cToF.isSelected()) {
                    double f = (temp * 9 / 5) + 32;
                    result.setText("Result: " + temp + " C = " + String.format("%.2f", f) + " F");
                } else {
                    double c = (temp - 32) * 5 / 9;
                    result.setText("Result: " + temp + " F = " + String.format("%.2f", c) + " C");
                }
            } catch (NumberFormatException ex) {
                JOptionPane.showMessageDialog(this, "Invalid input! Please enter a valid number.",
                        "Error", JOptionPane.ERROR_MESSAGE);
                result.setText("Result: ");
            }
        } else if (e.getSource() == clear) {
            tTemp.setText("");
            result.setText("Result: ");
            cToF.setSelected(true);
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Temperature = 100, "Celsius to Fahrenheit" selected, click Convert
Result: 100.0 C = 212.00 F

Temperature = 98.6, "Fahrenheit to Celsius" selected, click Convert
Result: 98.6 F = 37.00 C

Temperature = abc, click Convert
JOptionPane (Error): Invalid input! Please enter a valid number.

On clicking Clear: the field and result are cleared and
"Celsius to Fahrenheit" is selected again.
```

Q8. Simple Notepad - Swing

Create a basic Swing Notepad with JTextArea and a menu bar containing File -> New, Open, Save and Exit. New clears the text area; Open selects a text file; Save writes the contents to a file; Exit closes the application.

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.io.*;

public class Main extends JFrame implements ActionListener {
    JTextArea textArea;
    JMenuBar menuBar;
    JMenu fileMenu;
    JMenuItem newItem, openItem, saveItem, exitItem;

    Main() {
        setTitle("Simple Notepad");

        textArea = new JTextArea();
        textArea.setFont(new Font("Monospaced", Font.PLAIN, 14));
        add(new JScrollPane(textArea), BorderLayout.CENTER);

        menuBar = new JMenuBar();
        fileMenu = new JMenu("File");
        newItem = new JMenuItem("New");
        openItem = new JMenuItem("Open");
        saveItem = new JMenuItem("Save");
        exitItem = new JMenuItem("Exit");

        fileMenu.add(newItem);
        fileMenu.add(openItem);
        fileMenu.add(saveItem);
        fileMenu.addSeparator();
        fileMenu.add(exitItem);
        menuBar.add(fileMenu);
        setJMenuBar(menuBar);

        newItem.addActionListener(this);
```

```
        openItem.addActionListener(this);
        saveItem.addActionListener(this);
        exitItem.addActionListener(this);

        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(600, 400);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        JFileChooser chooser = new JFileChooser();

        if (e.getSource() == newItem) {
            textArea.setText("");
        } else if (e.getSource() == openItem) {
            if (chooser.showOpenDialog(this) == JFileChooser.APPROVE_OPTION) {
                File file = chooser.getSelectedFile();
                try {
                    BufferedReader br = new BufferedReader(new FileReader(file));
                    textArea.read(br, null);
                    br.close();
                } catch (IOException ex) {
                    JOptionPane.showMessageDialog(this, "Error opening file: " + ex.getMessage());
                }
            }
        } else if (e.getSource() == saveItem) {
            if (chooser.showSaveDialog(this) == JFileChooser.APPROVE_OPTION) {
                File file = chooser.getSelectedFile();
                try {
                    FileWriter fw = new FileWriter(file);
                    textArea.write(fw);
                    fw.close();
                    JOptionPane.showMessageDialog(this, "File saved successfully.");
                } catch (IOException ex) {
                    JOptionPane.showMessageDialog(this, "Error saving file: " + ex.getMessage());
                }
            }
        } else if (e.getSource() == exitItem) {
            System.exit(0);
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Menu bar:  File -> New | Open | Save | ---- | Exit

Typing in the text area:
Hello, this is my Java Notepad.
File handling with Swing is fun!
- Varnika

File -> New   : the text area is cleared.
File -> Open  : a JFileChooser opens; the selected text file is loaded
                into the text area.
File -> Save  : a JFileChooser opens; the contents are written to the
                chosen file and "File saved successfully." is shown.
File -> Exit  : the application closes.
```

Q9. Student Feedback Form - Swing

Create a Swing Student Feedback Form with Student Name, Roll Number, Course, rating radio buttons (Excellent, Very Good, Good, Average, Poor), feedback checkboxes (Teaching, Laboratory, Course Material, Assignments), Comments JTextArea, Submit and Reset. Submit should display all feedback using JOptionPane.

```
import javax.swing.*;
import javax.swing.border.*;
import java.awt.*;
import java.awt.event.*;

public class Main extends JFrame implements ActionListener {
    JTextField tName, tRoll, tCourse;
    JRadioButton[] rating = new JRadioButton[5];
    ButtonGroup ratingGroup;
```

```
JCheckBox[] aspects = new JCheckBox[4];
JTextArea comments;
JButton submit, reset;

Main() {
    setTitle("Student Feedback Form");
    setLayout(new BorderLayout(10, 10));

    // Top: student details
    JPanel top = new JPanel(new GridLayout(3, 2, 10, 10));
    top.setBorder(new EmptyBorder(15, 20, 5, 20));
    tName = new JTextField();
    tRoll = new JTextField();
    tCourse = new JTextField();
    top.add(new JLabel("Student Name:")); top.add(tName);
    top.add(new JLabel("Roll Number:"));  top.add(tRoll);
    top.add(new JLabel("Course:"));       top.add(tCourse);

    // Middle: rating and feedback options
    JPanel ratingPanel = new JPanel(new GridLayout(5, 1));
    ratingPanel.setBorder(new TitledBorder("Rating"));
    String[] r = {"Excellent", "Very Good", "Good", "Average", "Poor"};
    ratingGroup = new ButtonGroup();
    for (int i = 0; i < 5; i++) {
        rating[i] = new JRadioButton(r[i]);
        ratingGroup.add(rating[i]);
        ratingPanel.add(rating[i]);
    }

    JPanel aspectPanel = new JPanel(new GridLayout(4, 1));
    aspectPanel.setBorder(new TitledBorder("Feedback On"));
    String[] a = {"Teaching", "Laboratory", "Course Material", "Assignments"};
    for (int i = 0; i < 4; i++) {
        aspects[i] = new JCheckBox(a[i]);
        aspectPanel.add(aspects[i]);
    }

    JPanel options = new JPanel(new GridLayout(1, 2, 10, 10));
    options.setBorder(new EmptyBorder(0, 20, 0, 20));
    options.add(ratingPanel);
    options.add(aspectPanel);

    comments = new JTextArea(4, 20);
    JScrollPane sp = new JScrollPane(comments);
    sp.setBorder(new TitledBorder("Comments"));
    JPanel commentPanel = new JPanel(new BorderLayout());
    commentPanel.setBorder(new EmptyBorder(0, 20, 0, 20));
    commentPanel.add(sp);

    JPanel center = new JPanel(new BorderLayout(5, 5));
    center.add(options, BorderLayout.NORTH);
    center.add(commentPanel, BorderLayout.CENTER);

    // Bottom: buttons
    submit = new JButton("Submit");
    reset = new JButton("Reset");
    JPanel bottom = new JPanel(new FlowLayout());
    bottom.add(submit);
    bottom.add(reset);

    submit.addActionListener(this);
    reset.addActionListener(this);

    add(top, BorderLayout.NORTH);
    add(center, BorderLayout.CENTER);
    add(bottom, BorderLayout.SOUTH);

    setDefaultCloseOperation(EXIT_ON_CLOSE);
    setSize(470, 520);
    setLocationRelativeTo(null);
    setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == submit) {
        String rate = "Not selected";
        for (JRadioButton rb : rating) {
            if (rb.isSelected()) rate = rb.getText();
        }
```

```
            String areas = "";
            for (JCheckBox cb : aspects) {
                if (cb.isSelected()) areas += cb.getText() + ", ";
            }
            areas = areas.isEmpty() ? "None" : areas.substring(0, areas.length() - 2);

            JOptionPane.showMessageDialog(this,
                    "Student Name : " + tName.getText() + "\n"
                    + "Roll Number  : " + tRoll.getText() + "\n"
                    + "Course       : " + tCourse.getText() + "\n"
                    + "Rating       : " + rate + "\n"
                    + "Feedback On  : " + areas + "\n"
                    + "Comments     : " + comments.getText(),
                    "Feedback Submitted", JOptionPane.INFORMATION_MESSAGE);
        } else if (e.getSource() == reset) {
            tName.setText("");
            tRoll.setText("");
            tCourse.setText("");
            ratingGroup.clearSelection();
            for (JCheckBox cb : aspects) cb.setSelected(false);
            comments.setText("");
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Input: Name = Varnika, Roll Number = CU25260192, Course = BCA,
Rating = Very Good, Feedback On = Teaching, Laboratory,
Comments = Labs are very helpful. Please add more practice sheets.

On clicking Submit, JOptionPane shows (title: Feedback Submitted):
Student Name : Varnika
Roll Number  : CU25260192
Course       : BCA
Rating       : Very Good
Feedback On  : Teaching, Laboratory
Comments     : Labs are very helpful.
Please add more practice sheets.

On clicking Reset: all fields, rating, checkboxes and comments are cleared.
```

Q10. Mini Banking GUI Application - Swing

Create a Swing Mini Banking Application with Account Number, Customer Name, Current Balance, Amount and buttons Deposit, Withdraw, Check Balance and Clear. Deposit adds amount; Withdraw subtracts it only when sufficient balance exists; Check Balance displays balance; Clear clears inputs. Use JOptionPane for messages.

```
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;
import java.awt.event.*;

public class Main extends JFrame implements ActionListener {
    JTextField tAccNo, tName, tBalance, tAmount;
    JButton deposit, withdraw, check, clear;
    double balance = 5000.00;   // opening balance

    Main() {
        setTitle("Mini Banking Application");
        setLayout(new BorderLayout(10, 10));

        JPanel form = new JPanel(new GridLayout(4, 2, 10, 12));
        form.setBorder(new EmptyBorder(20, 20, 5, 20));

        tAccNo = new JTextField();
        tName = new JTextField();
        tBalance = new JTextField(String.format("%.2f", balance));
        tBalance.setEditable(false);
        tAmount = new JTextField();

        form.add(new JLabel("Account Number:")); form.add(tAccNo);
        form.add(new JLabel("Customer Name:"));  form.add(tName);
        form.add(new JLabel("Current Balance:")); form.add(tBalance);
        form.add(new JLabel("Amount:"));         form.add(tAmount);
        deposit = new JButton("Deposit");
```

```
        withdraw = new JButton("Withdraw");
        check = new JButton("Check Balance");
        clear = new JButton("Clear");
        JPanel buttons = new JPanel(new GridLayout(1, 4, 8, 8));
        buttons.setBorder(new EmptyBorder(5, 20, 20, 20));
        buttons.add(deposit);
        buttons.add(withdraw);
        buttons.add(check);
        buttons.add(clear);

        deposit.addActionListener(this);
        withdraw.addActionListener(this);
        check.addActionListener(this);
        clear.addActionListener(this);

        add(form, BorderLayout.CENTER);
        add(buttons, BorderLayout.SOUTH);

        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(520, 250);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    double readAmount() {
        try {
            double amt = Double.parseDouble(tAmount.getText().trim());
            if (amt <= 0) {
                JOptionPane.showMessageDialog(this, "Amount must be greater than zero.",
                        "Invalid Amount", JOptionPane.WARNING_MESSAGE);
                return -1;
            }
            return amt;
        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(this, "Please enter a valid amount.",
                    "Invalid Input", JOptionPane.ERROR_MESSAGE);
            return -1;
        }
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == deposit) {
            double amt = readAmount();
            if (amt > 0) {
                balance += amt;
                tBalance.setText(String.format("%.2f", balance));
                JOptionPane.showMessageDialog(this, "Rs. " + amt + " deposited successfully.");
            }
        } else if (e.getSource() == withdraw) {
            double amt = readAmount();
            if (amt > 0) {
                if (amt <= balance) {
                    balance -= amt;
                    tBalance.setText(String.format("%.2f", balance));
                    JOptionPane.showMessageDialog(this, "Rs. " + amt + " withdrawn successfully.");
                } else {
                    JOptionPane.showMessageDialog(this, "Insufficient balance!",
                            "Error", JOptionPane.ERROR_MESSAGE);
                }
            }
        } else if (e.getSource() == check) {
            JOptionPane.showMessageDialog(this, "Current Balance: Rs. " + String.format("%.2f", balance));
        } else if (e.getSource() == clear) {
            tAccNo.setText("");
            tName.setText("");
            tAmount.setText("");
        }
    }

    public static void main(String[] args) {
        new Main();
    }
}
```

OUTPUT:-

```
Opening balance: 5000.00
Account Number = 1001, Customer Name = Varnika

Amount = 2000, click Deposit
JOptionPane: Rs. 2000.0 deposited successfully.   (Balance: 7000.00)

Amount = 10000, click Withdraw
JOptionPane (Error): Insufficient balance!        (Balance: 7000.00)

Amount = 1500, click Withdraw
JOptionPane: Rs. 1500.0 withdrawn successfully.   (Balance: 5500.00)

Click Check Balance
JOptionPane: Current Balance: Rs. 5500.00

Invalid / negative amount -> "Please enter a valid amount." /
"Amount must be greater than zero."
On clicking Clear: account number, name and amount are cleared.
```